Senior Health Workforce Analyst - Assignment

Health Workforce Modelling

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About me

Applied quantitative modeler with app development experience.

- Training:
  - PhD. in Systems Design Engineering
  - MSc. in Operations Research
  - BSc. in Industrial Management

- My modelling experience:
  - Decision models (optimization)
  - Agent-based simulation
  - Physics-based modelling
  - Macroeconomic models

Goal:

- Apply quantitative modeling and data analytics to improve well-being of Ontarians.
- Learn and grow as a professional by taking on new challenges and responsibilities.

Assignment & Modelling Approach

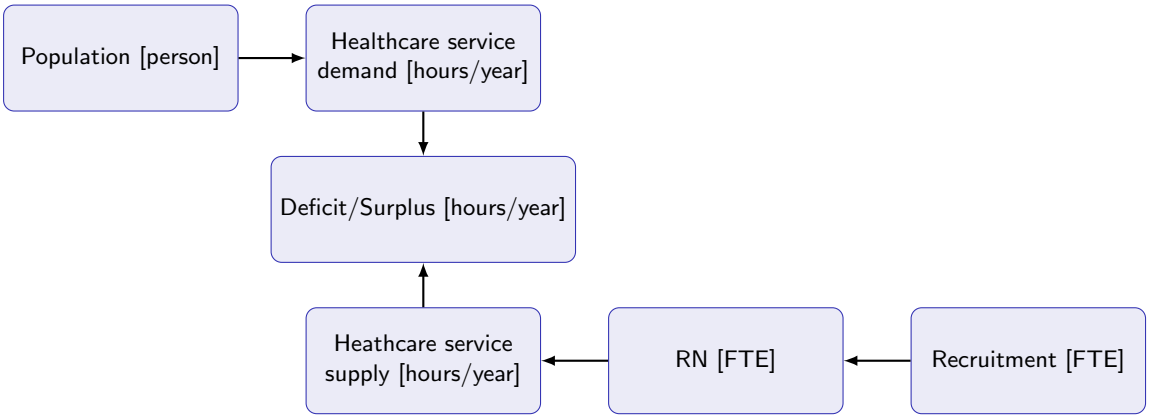
Develop a quantitative model to forecast workforce supply and workforce requirements for **Registered Nurses** (RNs) in Ontario over the next 10 years.

Modelling approach:

- Start simple and enrich only if needed.
- Research or policy question being addressed (how a solution looks like?).
- Decision variables or policy instruments (what can be controlled?).
- Causal relationships between variables.

There is no good solution to a badly posed problem.

Conceptual formulation



- FTE: Full-time equivalent.

Model Decisions

- Which variables are considered exogenous?
- Spatial and temporal granularity.
- Most appropriate model type, e.g.:
  - System dynamics
  - Time series
  - Optimization
  - Agent-based modelling
- Model should drive data sources (rather than the other way around).
- Look for solutions in literature.
- Start with a simple model first.

Demand-side Modelling

- Demand of healthcare services: d_t [hours/time]
- α: per capita demand for healthcare services [hours/(person time)]
- Population-driven demand:

$$d_t = \alpha \; p_t \quad \forall t \qquad \text{[hours/time]}$$

- Age-based demand: $d_t = \sum_a \alpha_a \; p_{a,t}$
- Specialty-based demand: $d_{t,s} = \sum_a \alpha_{a,s} \; p_{a,t}$

Population Modelling

- Leslie (age-structured) model, in vector form:

$$\mathbf{p}_{t+1} = \mathbf{L}\ \mathbf{p}_t + \mathbf{i}_t \quad \forall t$$

- $\mathbf{p}_t$: population vector at time t.
- $\mathbf{L}$: Leslie matrix.
- $\mathbf{i}_t$: net immigration vector at time t (major source of uncertainty).

Supply-side Modelling

- Supply of healthcare services: s_t [hours/time]
- β_t: per capita supply of healthcare services [hours/(FTE time)]
- n_t: number of healthcare professionals (FTE) at time t.
- Supply of healthcare services:

$$s_t = \beta \; n_t \quad \forall t \qquad \text{[hours/time]}$$

- Specialty-based supply: $s_{t,s} = \beta_s \; n_{s,t}$

Balance Surplus/Deficit

- Balance of healthcare services:

$$y_t = d_t - s_t \quad \forall t \qquad \text{[hours/time]}$$

- Surplus: $y_t^+ = \max(y_t, 0)$
- Deficit: $y_t^- = \max(-y_t, 0)$

Optimal hiring policy model

Decision variable x_t: new RNs hired at time t.

$$\min_{x_t} \sum_{t=0}^{T} \lambda \; e^{-\delta t} \, y_t^+ + \sum_{t=0}^{T} \mu \; e^{-\delta t} \, y_t^-$$

s.t.

$$d_t = \alpha_t \; p_t \quad \forall t \tag{1}$$

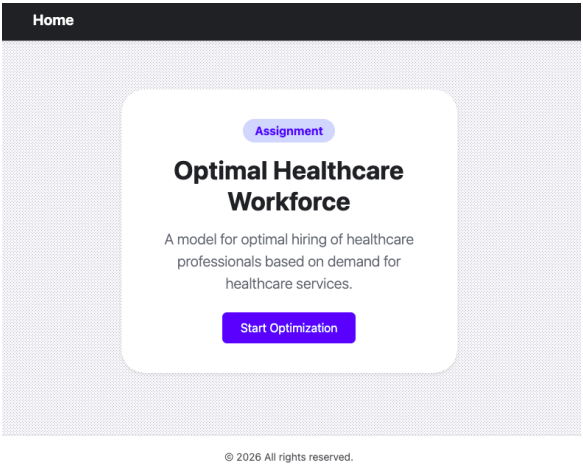
$$s_t = \beta_t \; n_t \quad \forall t \tag{2}$$

$$d_t - s_t = y_t^+ - y_t^- \quad \forall t \tag{3}$$

$$n_t = \gamma \; n_{t-1} + x_t \quad \forall t \tag{4}$$

$$x_t, y_t^+, y_t^- \geq 0 \quad \forall t \tag{5}$$

Decision Support Tool



Decision Support Tool: https://20.220.16.0/

Development tools

- Model formulation: OR-Tools
- Application framework: Flask
- Infrastructure:
  - Azure virtual machine (Ubuntu 24.04)
  - Nginx, Gunicorn
- Available at `https://20.220.16.0/`
- Project repository: `https://github.com/jorge-antares/healthworkforce`

Extensions

- Population's dynamics (Leslie model).
- Capacity constraints due to facilities: $n_t \leq c_t$
- Recruitment sources $x_{t,s}$:
  - domestic graduates
  - international educated nurses
  - re-entrants
- Budget constraints on wages or recruitment.
- Bring economic theory: price elasticity of supply of nurses.

Data sources

- Demographic data:
  - Ministry of Finance population projections
  - Statistics Canada
  - CIHI health service utilization
- Healthcare professionals data:
  - College of Nurses of Ontario registration statistics
  - CIHI Nursing in Canada
  - Program enrolment and graduation counts
- Other sources as needed by the model
- **Privacy:** PII not needed for an aggregated model
- **Data gap:**
  - Interpolation.
  - Proportionality assumptions in the absence of detailed data.
  - Use of proxy variables.

Uncertainty Analysis

Ways to address uncertainty:

1. Probability distributions for key parameters or equations.
   - Monte Carlo simulation.
   - Expectation and Covariance matrix.
   - Confidence intervals.
2. Error propagation analysis:
   - Bounds on outputs.
3. Scenario analysis:
   - Optimistic-case, pessimistic-case, and most-likely scenarios.
4. Sensitivity analysis:
   - Assessing the impact of changes in key parameters on model outputs.

Disclosure of Tools

Claude Sonnet 5.5 was used in the following tasks:

- Create HTML for user interface.
- Create initial presentation template.
- Code completion and debugging suggestions (VS Code).

All LLM outputs were reviewed and edited.
All model formulation and assumptions are 100% the author's own.

Summary

- Use population as main driver.
- Start small and add complexity as needed.
- Construct a model that finds the least-cost to address supply-demand gaps.
- Address uncertainty through probability, error propagation, scenarios or sensitivity analysis.

All models are wrong, but some are useful.

Thank you — questions?